```java
class LibraryCatalog{
    //...
    public void updateAuthorInfo(Book b, String info){
        try {
            b.author.setInfo(info);
        } catch (ReadOnlyBookException robe){
            System.out.println(">>>>Error: author info cannot be edited!");
        }
    }
}
```

Which principles is most clearly violated?

- ○ Principle of Separation of Concerns
- ○ Single Responsibility Principle
- ○ Liskov Substitution Principle
- ○ Open Closed Principle
- ○ Law of Demeter

Q1

```
1   class LibraryCatalog{
2       //...
3       public void updateAuthorInfo(Book b, String info){
4           try {
5               b.author.setInfo(info);
6           } catch (ReadOnlyBookException robe){
7               System.out.println(">>>>Error: author info cannot be edited!");
8           }
9       }
10  }
```

Which principles is most clearly violated?

Yes	Maybe	No	○ Principle of Separation of Concerns	Separate the code into distinct sections, such that each section addresses a separate concern.
Yes	Maybe	No	○ Single Responsibility Principle	A class should have one, and only one, reason to change.
Yes	Maybe	N/A	○ Liskov Substitution Principle	Derived classes must be substitutable for their base classes.
Yes	Maybe	N/A	○ Open Closed Principle	A module should be *open* for extension but *closed* for modification.
Yes	Maybe	No	○ Law of Demeter	Don't talk to strangers

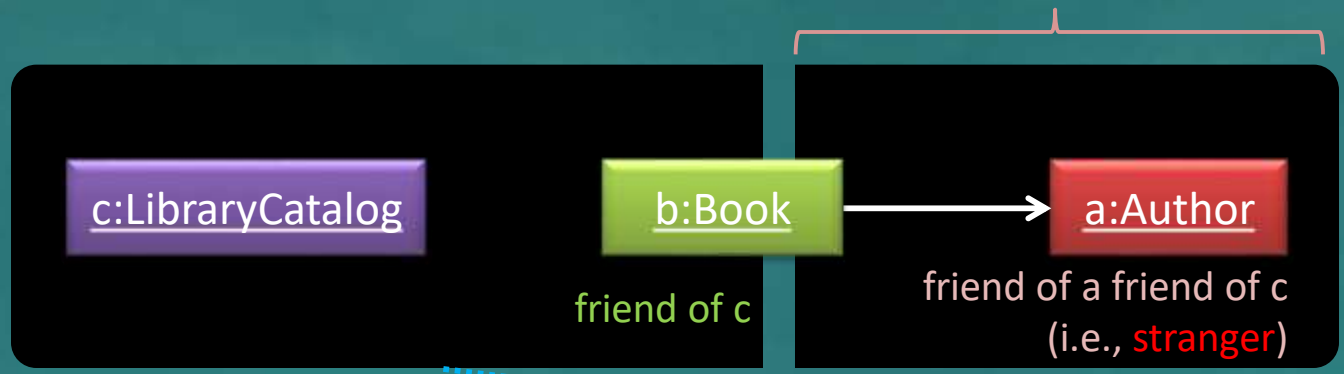
More on LoD violations …

```
1   class LibraryCatalog{
2       //...
3       public void updateAuthorInfo(Book b, String info){
4           try {
5               b.author.setInfo(info);
6           } catch (ReadOnlyBookException robe){
7               System.out.println(">>>>Error: author info cannot be edited!");
8           }
9       }
10  }
```

```
b.getAuthor().setInfo();  // same problem (but slightly better)!
```

☹ LibraryCatalog has a dependency on the internal object structure of Book!
What if we wanted to move the author object to a different location?

i.e., Do not navigate inside the internal object structure!



When paying a delivery person, do we allow that person to reach into your pocket, take your wallet out, and take money from it?
NO !!!

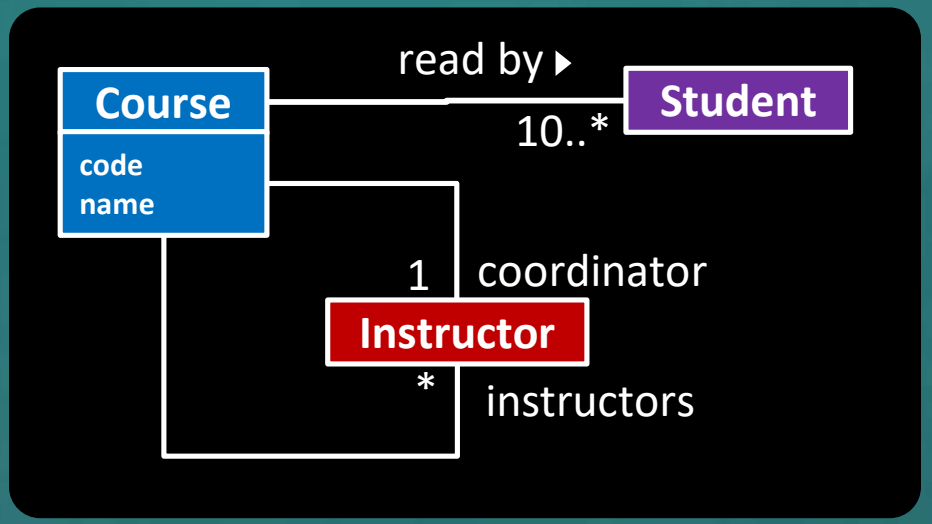
Alternatives:

- Receive an Author object instead of a Book
- Ask the Book to change author info.

Perhaps OK, as the method name clearly states it. But try to minimize.

Don't talk to strangers

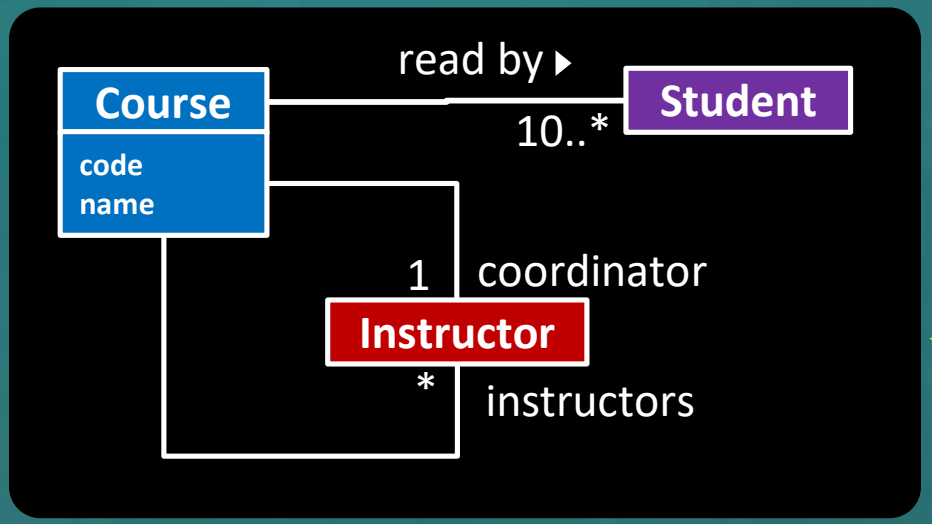
Q2 Which statements about this CCD are correct?



CCD is a model of the real world.
The design and the implementation of the solution can be guided by it, but it's not the same.
→ **avoid implementation-specific terminology** when discussing the problem domain.

- ❑ A. The `instructors` variable in the `Course` class can be of type `ArrayList<Instructor>`
- ❑ B. The `Course` class can have methods to add students to a specific course.
- ❑ C. There are alternative designs that can be more efficient to implement.
- ❑ D. Although not shown in the diagram, it is likely the `Student` class has attributes such as name, student number, gender etc.
- ❑ E. Instructor objects can play the *role* of coordinator.

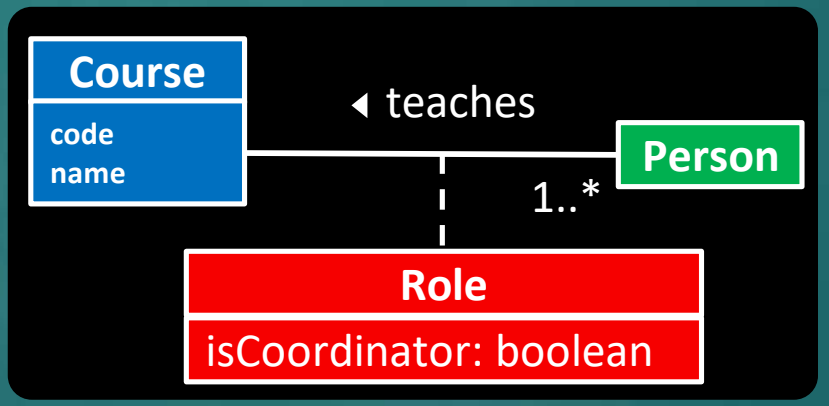
A course has a name and a code. A course is read by 10 or more students, and taught by one or more instructors one of whom is the coordinator.

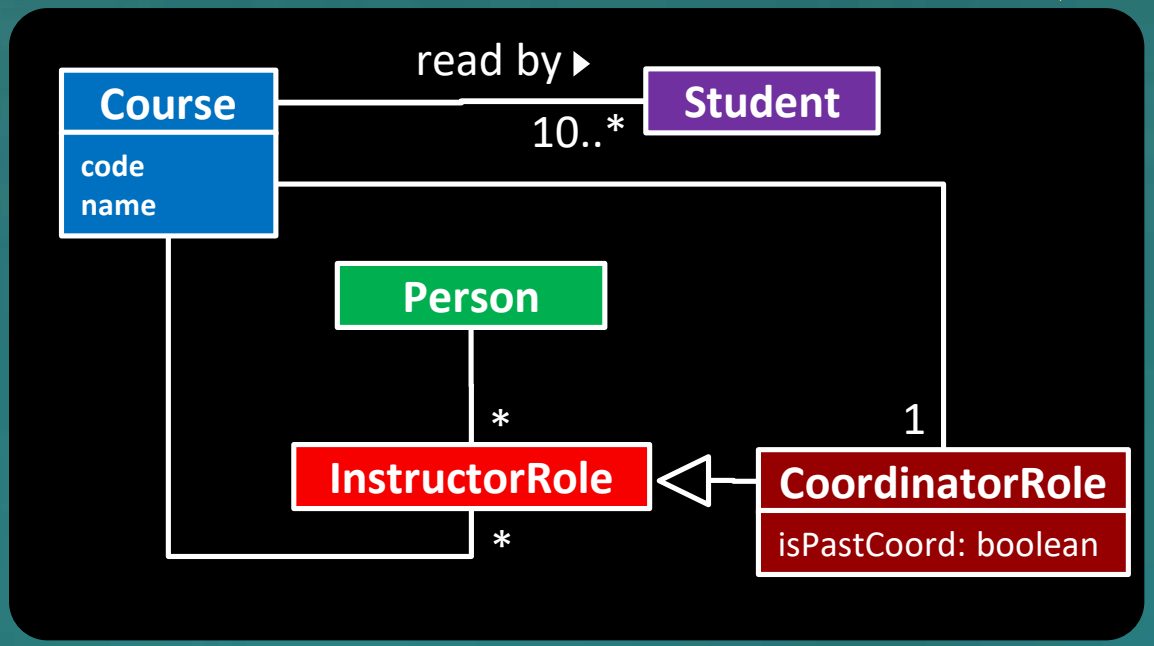


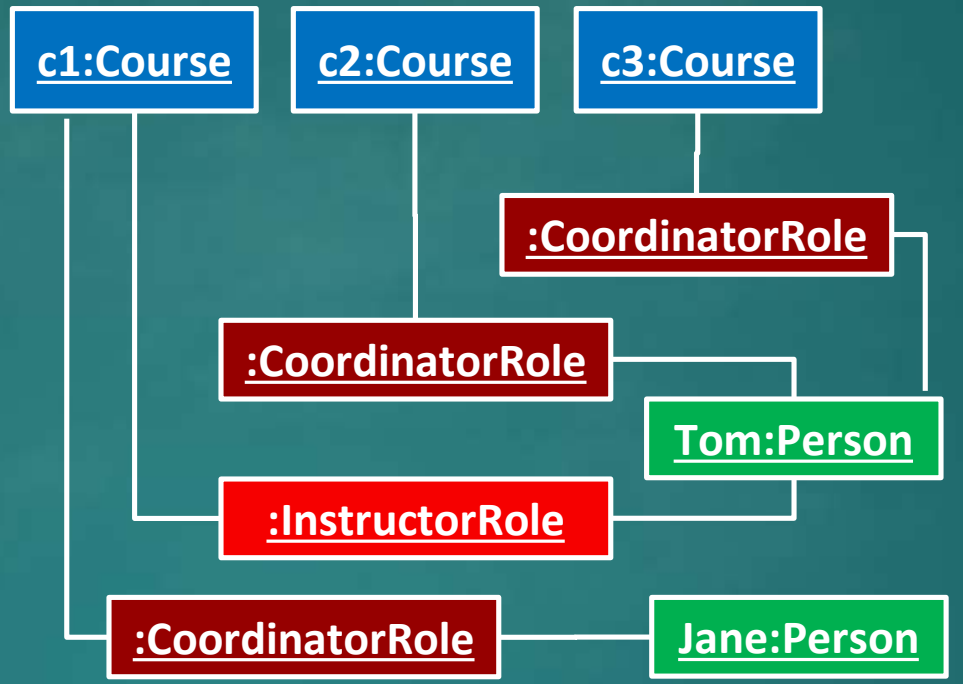
Multiple acceptable ways to model a domain!

Simpler

More flexible



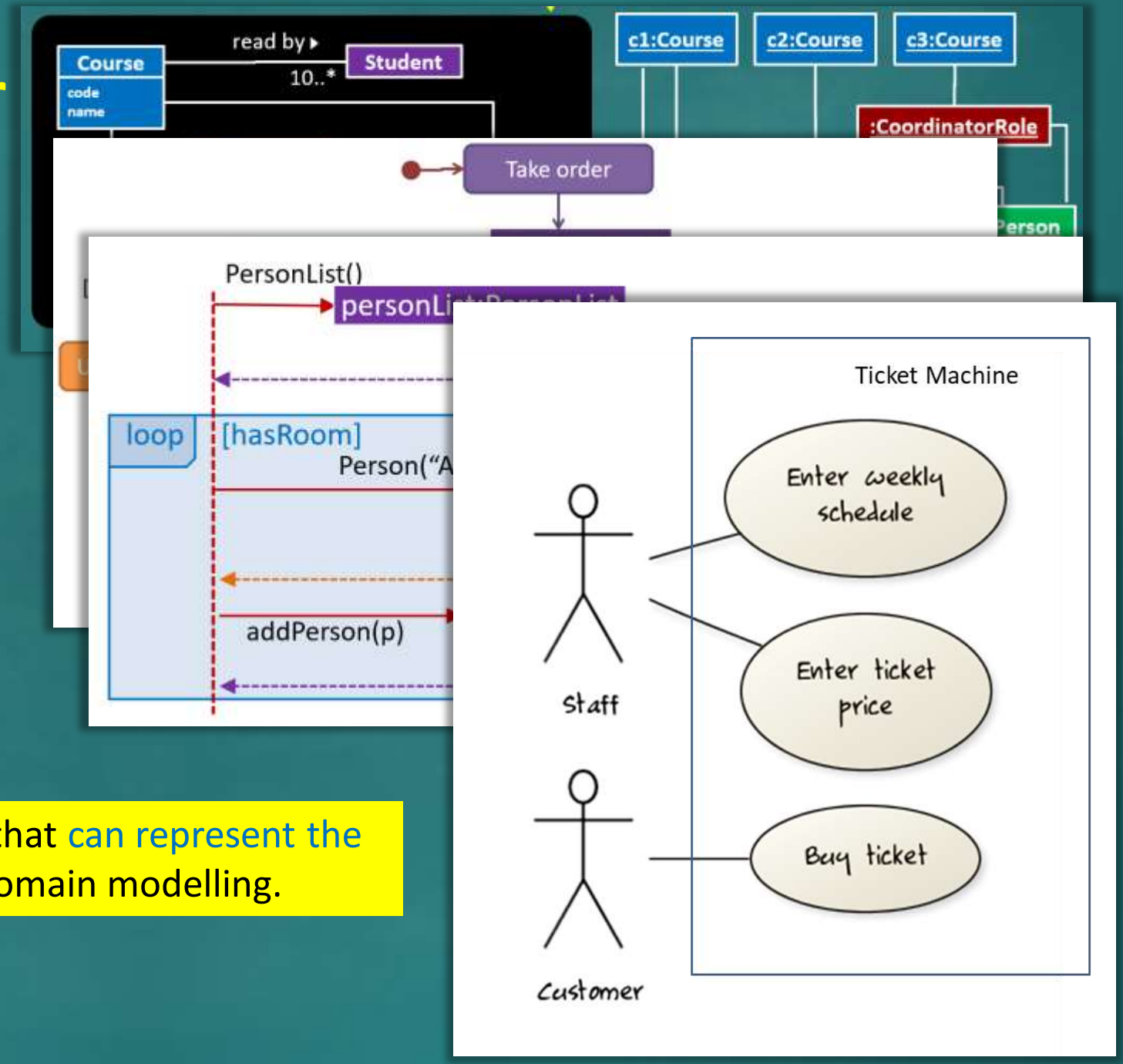




Q3

Which can be used for *domain* modelling?

- ❑ A CCD
- ❑ An object diagram
- ❑ An activity diagram
- ❑ A sequence diagram
- ❑ A class diagram
- ❑ A use case diagram



Anything (not just UML diagrams) that can represent the problem domain can be used for domain modelling.

Q4

Which SDLC process model is used by Google?

- ○ Scrum
- ○ XP
- ○ Unified process
- ○ None of the above

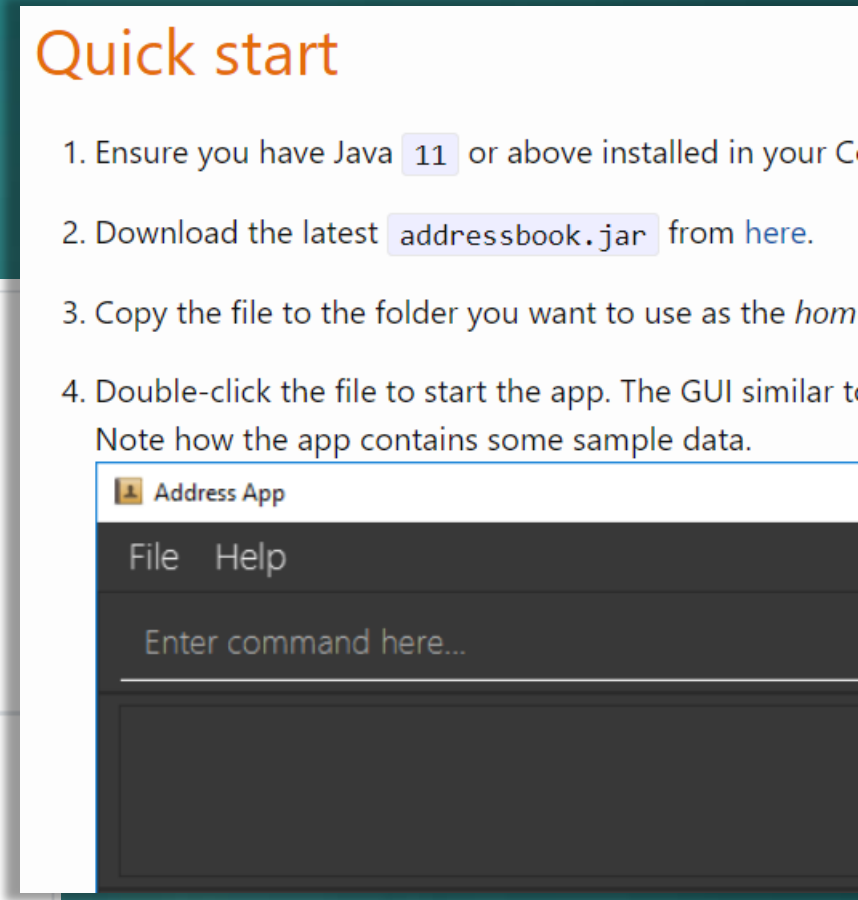
Some companies ‘claim’ to follow a specific process.

Many companies do use specific parts
e.g., *daily scrum*.

Most companies have their own process(es) that may borrow from these process models.

Q5 The ‘Quick start’ section of your tP UG belongs to which category?

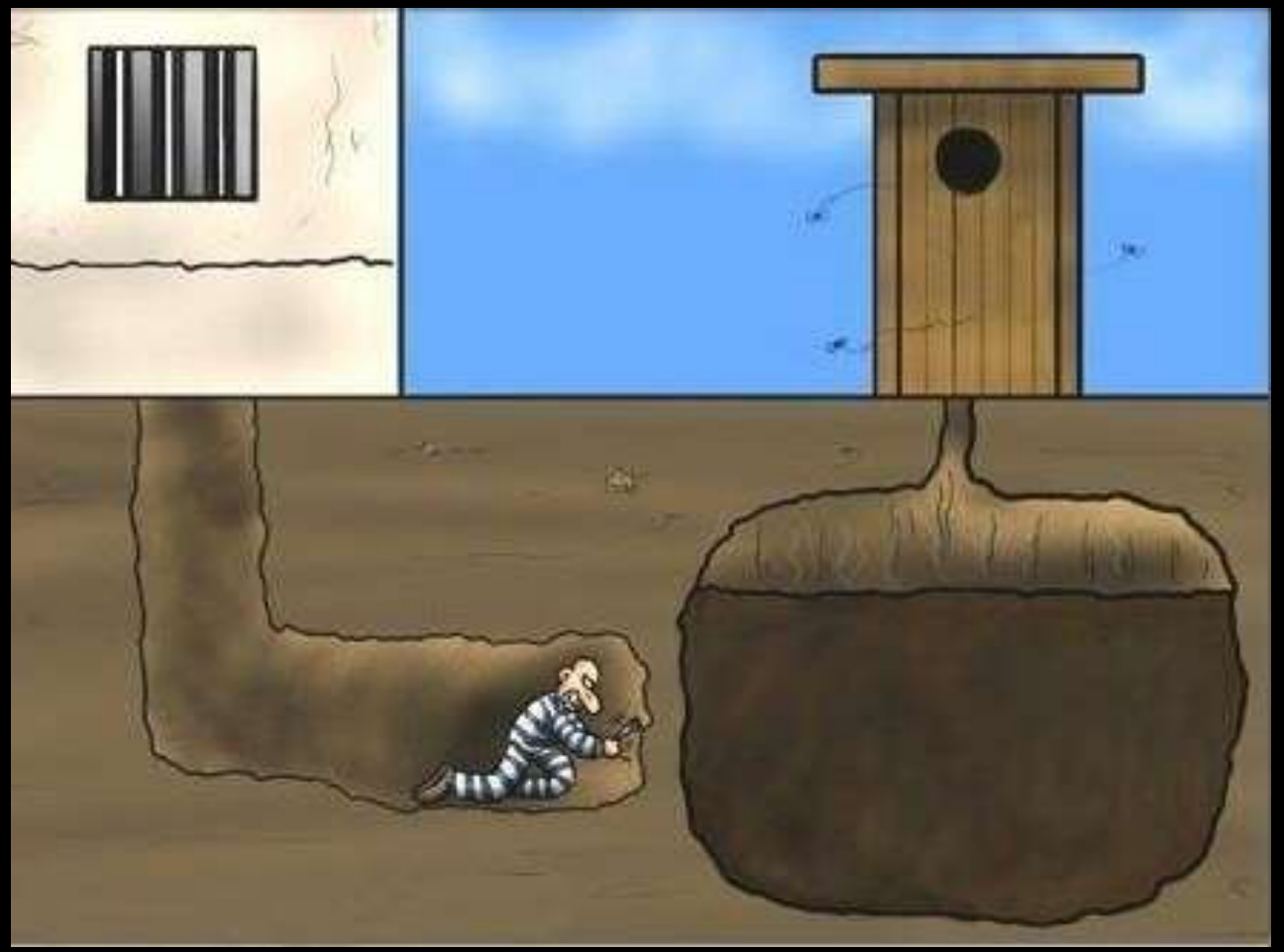
TUTORIALS	HOW-TO GUIDES
A tutorial: • is learning-oriented • allows the newcomer to get started • is a lesson Analogy: teaching a small child how to cook	A how-to guide: • is goal-oriented • shows how to solve a specific problem • is a series of steps Analogy: a recipe in a cookery book
EXPLANATION	**REFERENCE**
An explanation: • is understanding-oriented • explains • provides background and context Analogy: an article on culinary social history	A reference guide: • is information-oriented • describes the machinery • is accurate and complete Analogy: a reference encyclopedia article

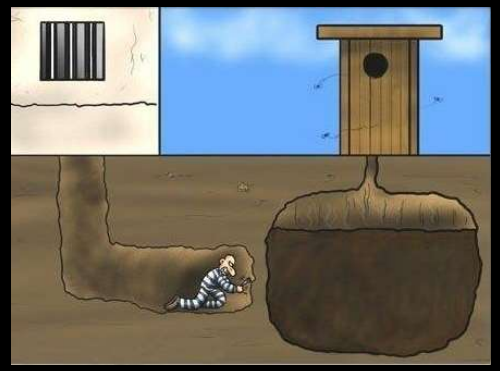


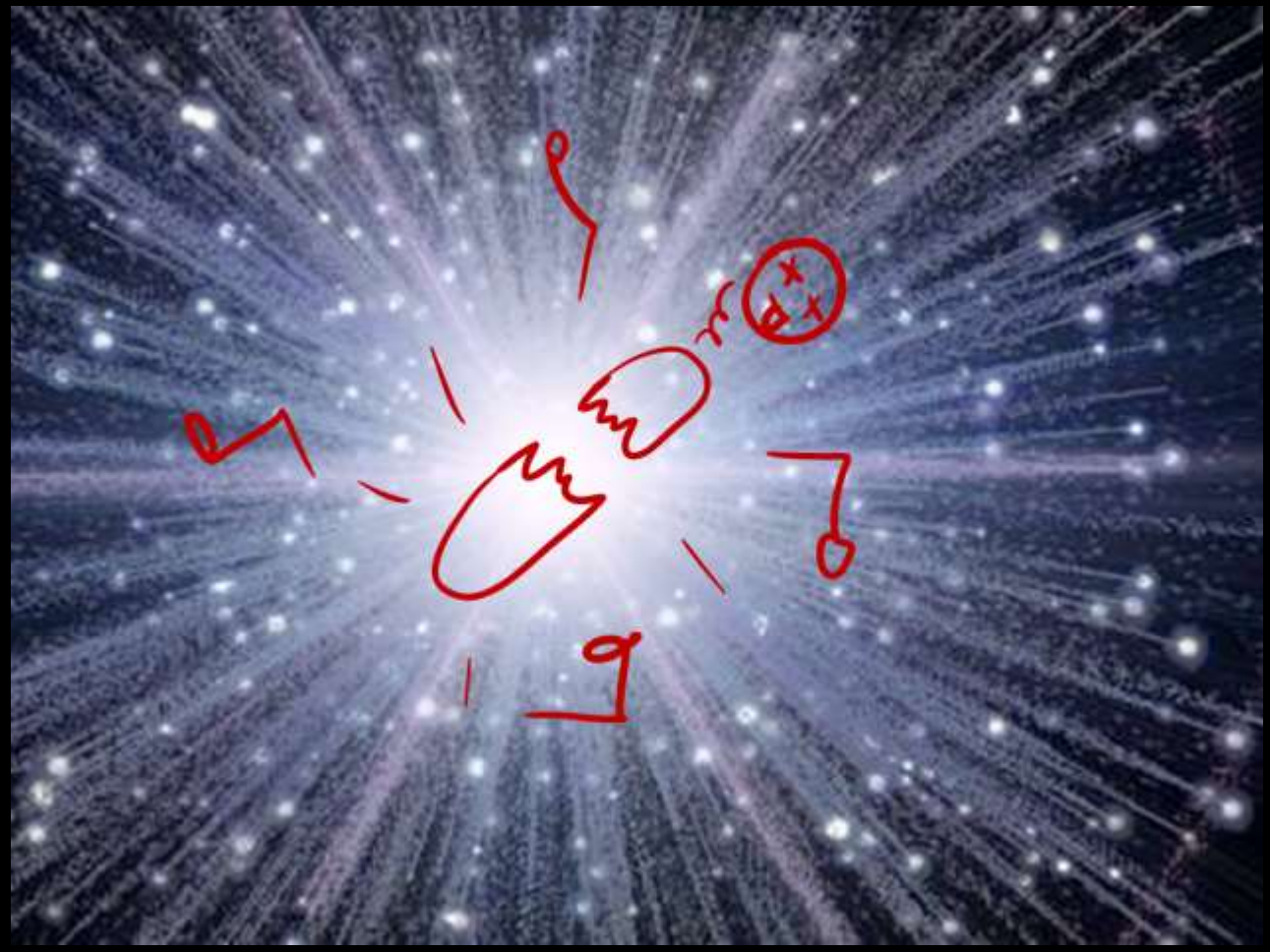
Try to keep each doc within one category,
although minor cross-overs may be unavoidable.

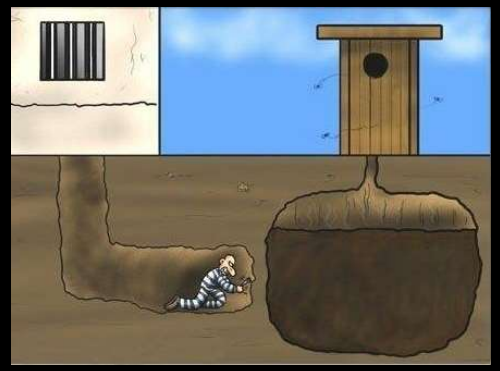
v1.3

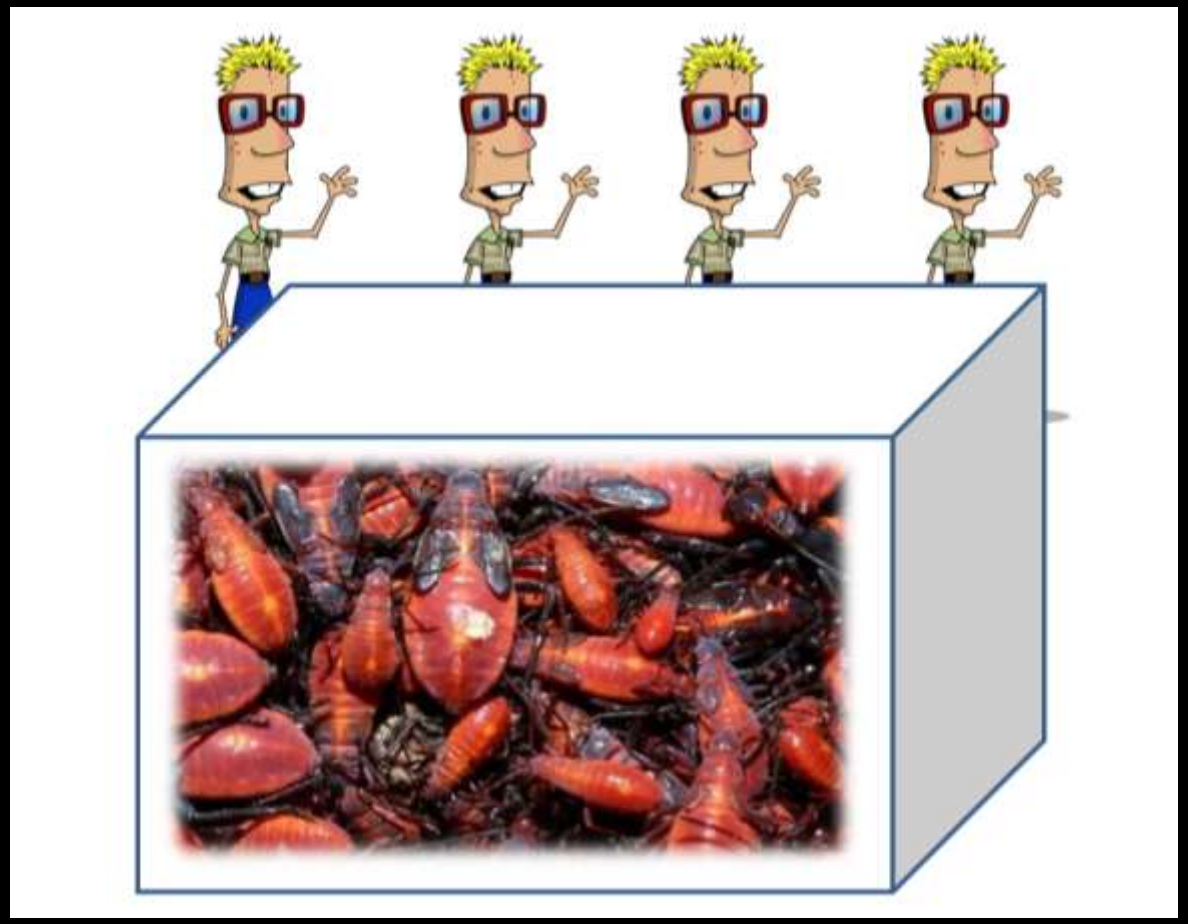
How?

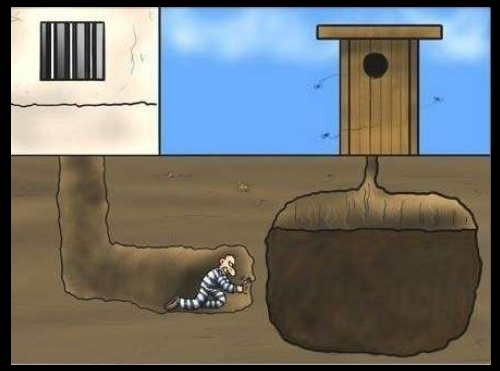


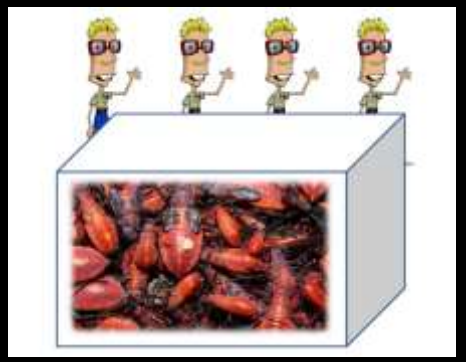


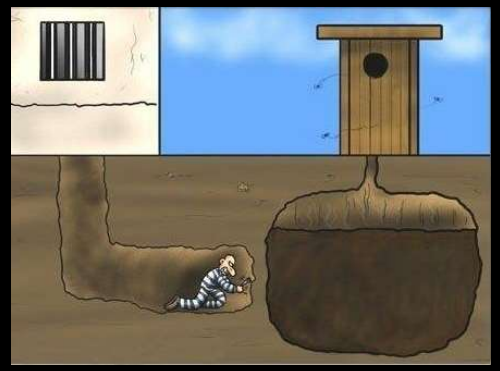


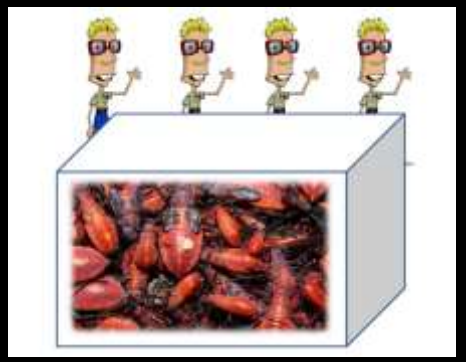




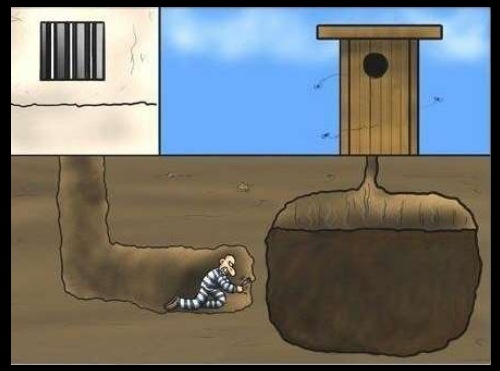


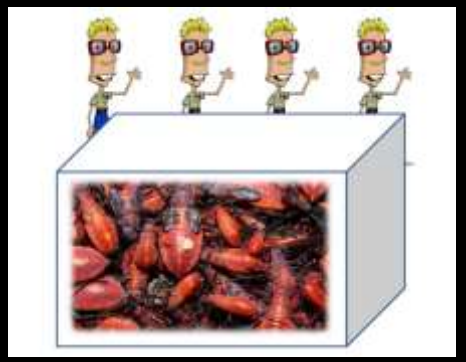






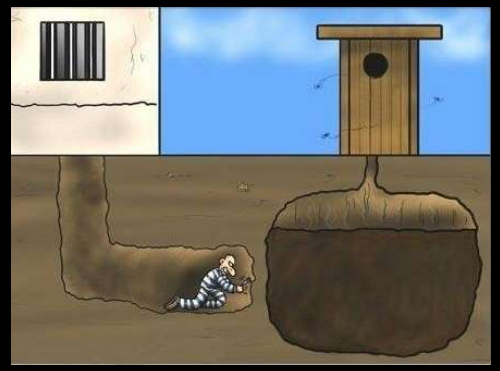

GBR

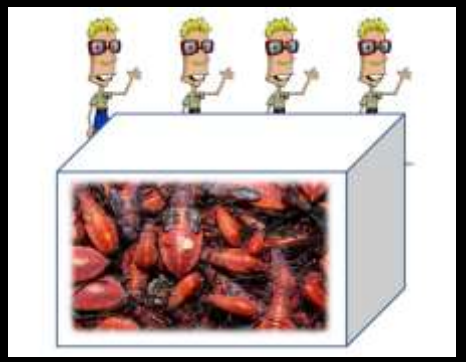





GBR

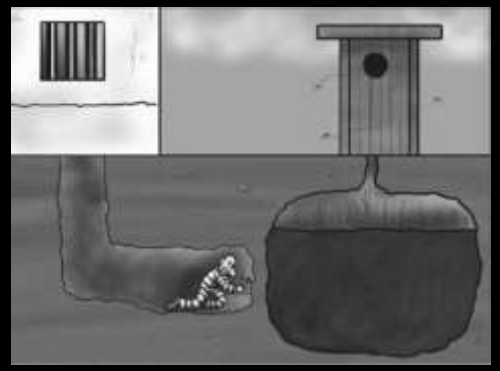

34
OSAKA

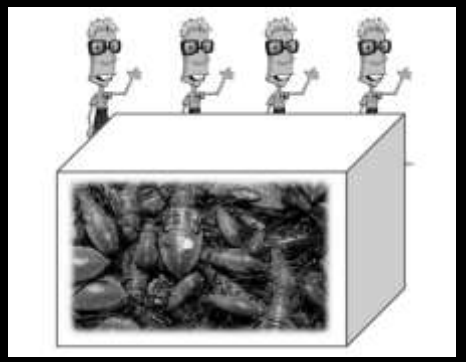




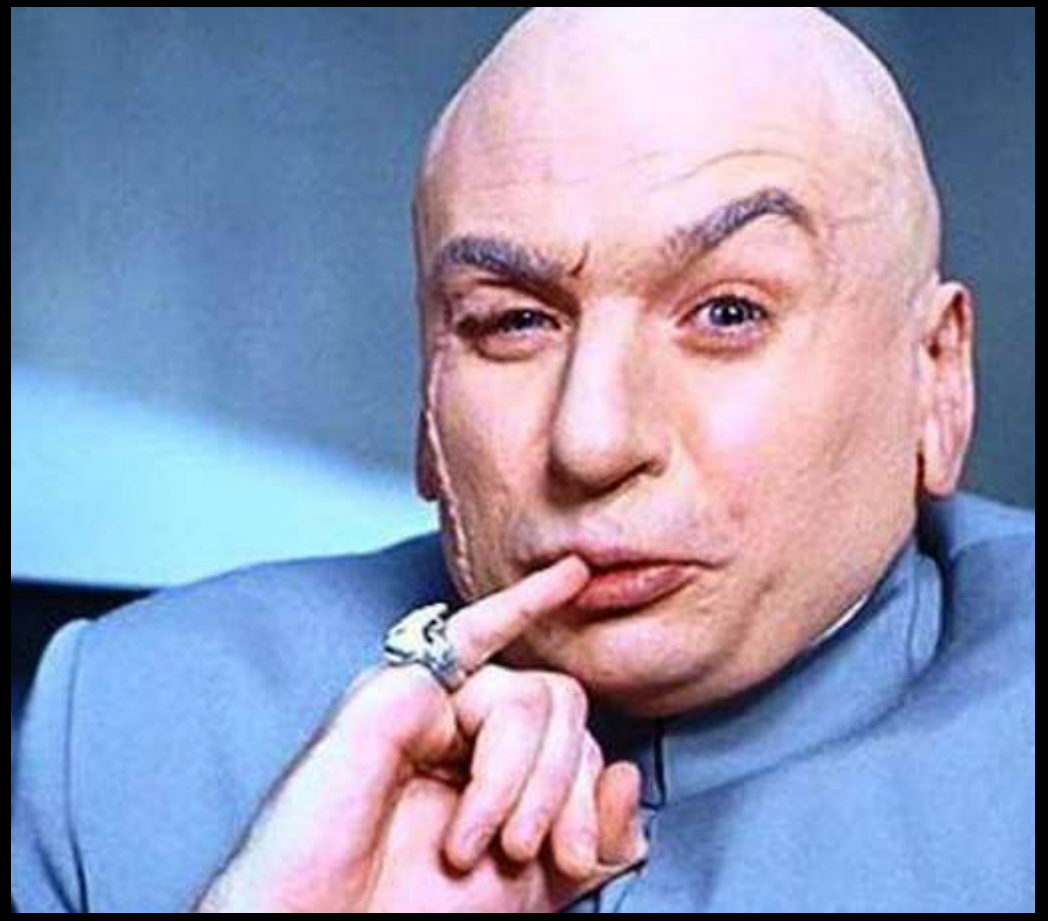

TK
34
OSAKA


Minesweeper
Game
Help

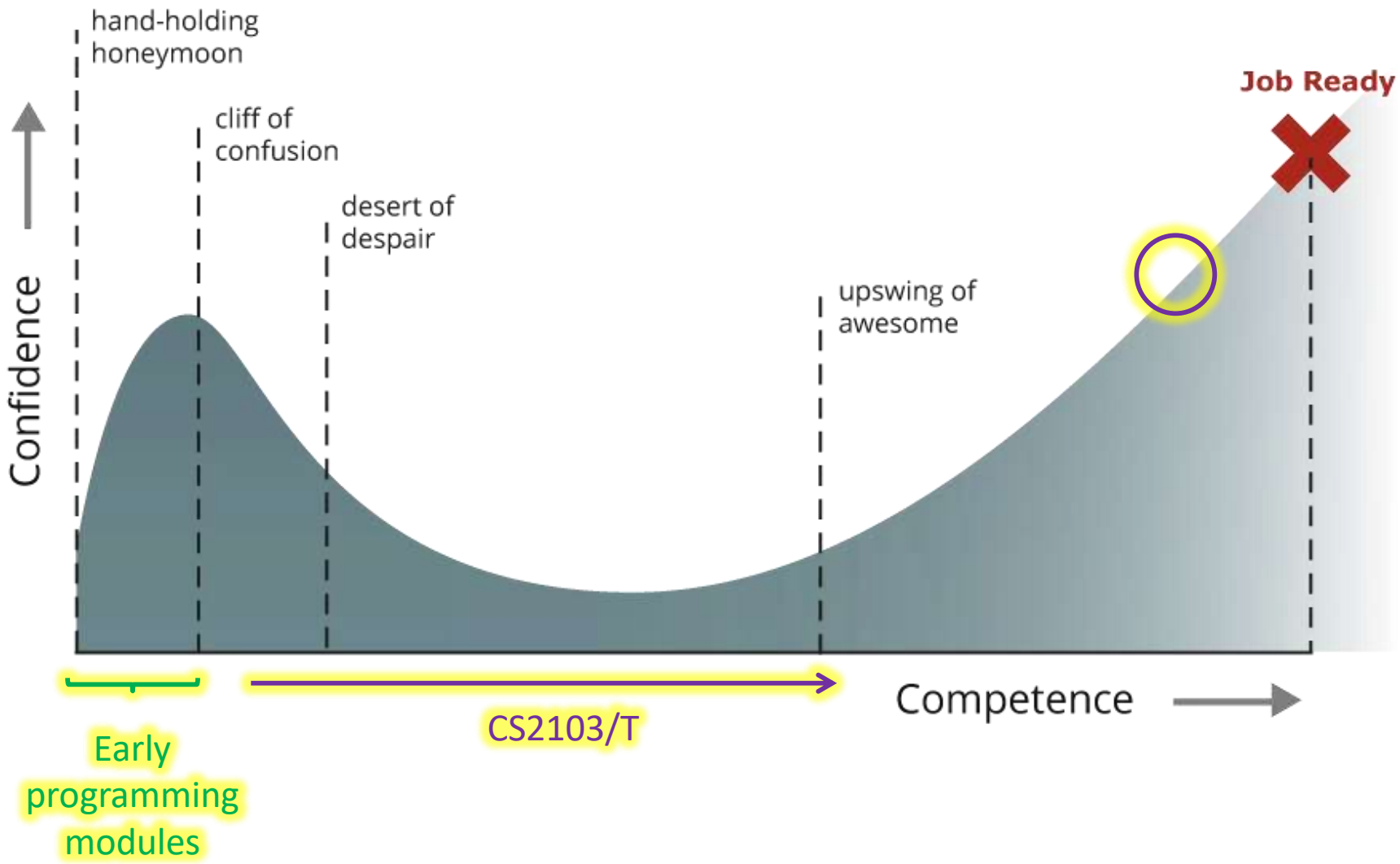





Minesweeper
Game
Help



Coding Confidence vs Competence



http://www.vikingcodeschool.com/posts/why-learning-to-code-is-so-damn-hard

v1.3

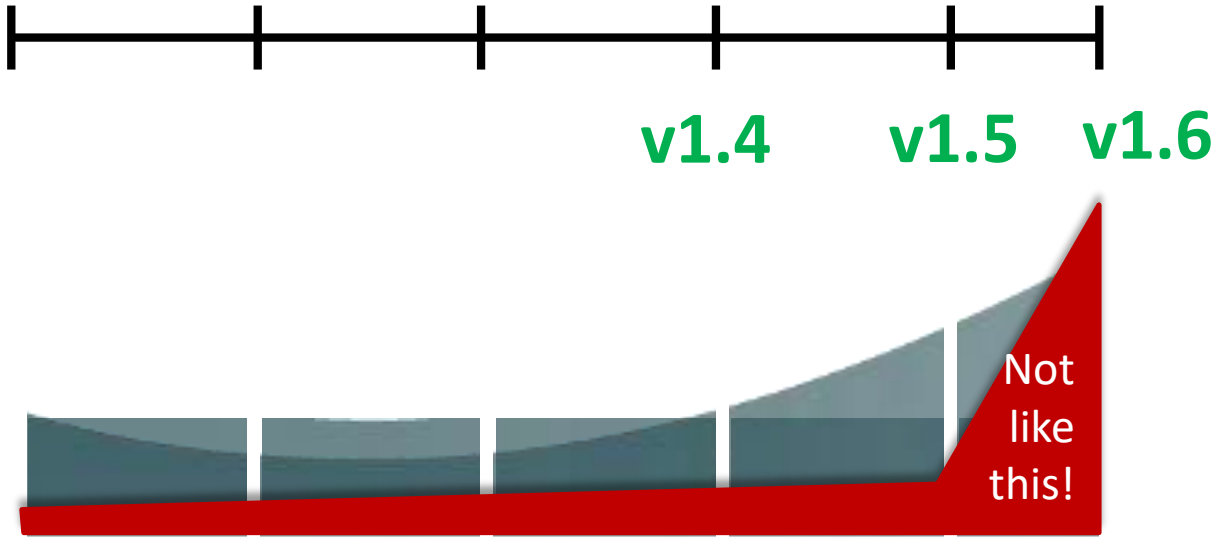
v1.6

v1.3

v1.6

v1.3

v1.6



v1.2

v1.6

1. Aim to create a gem you will be proud to wear

2. Polish, check, polish, check, polish, check,…

3. Sharpen your tools once in a while

Week 10

Topics:

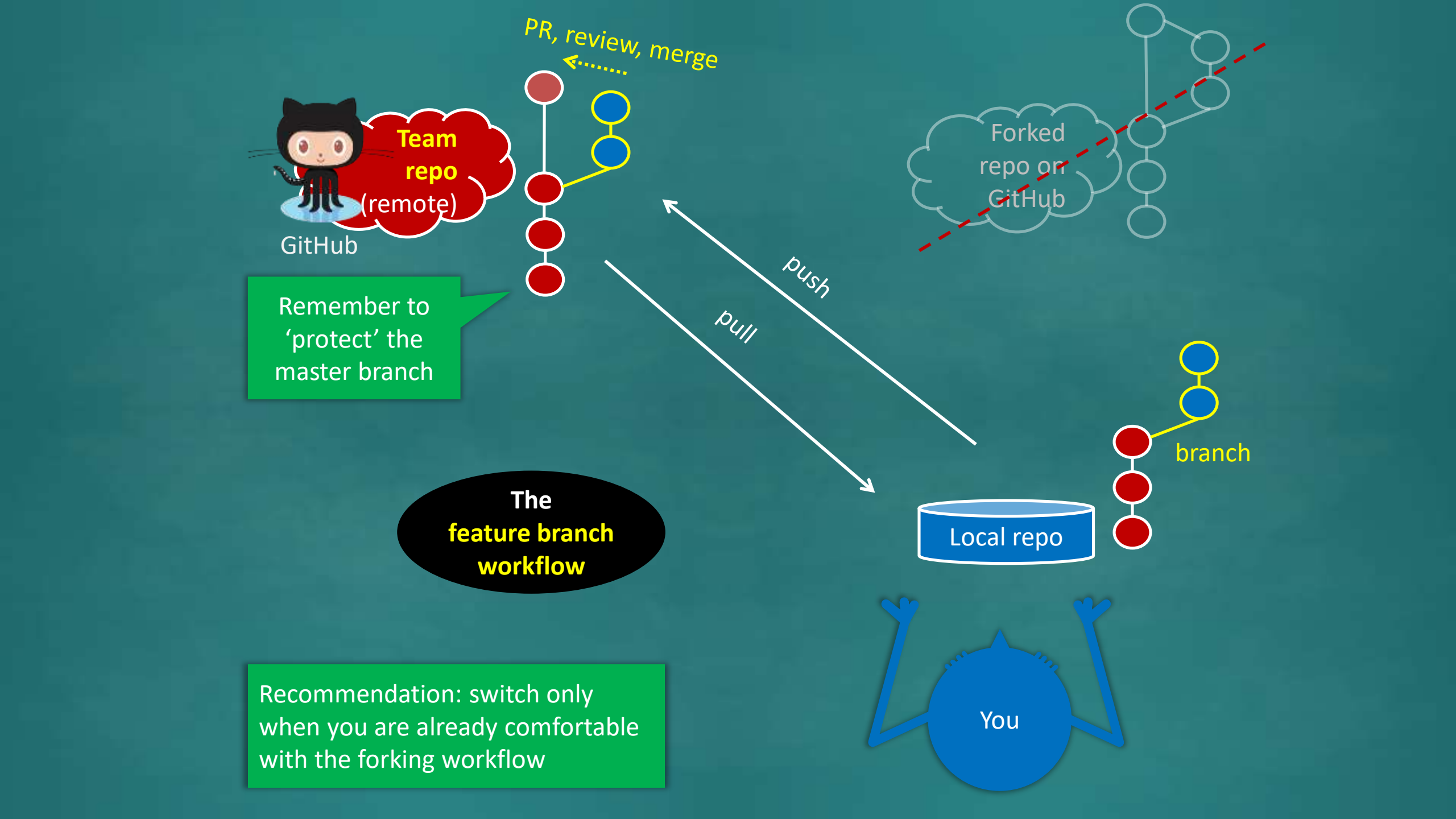
- [W10.1] **Design Patterns** ⚡⚡
- [W10.2] **Defensive Programming** ⚡⚡
- [W10.3] **Test Cases: Intro** ⚡⚡
- [W10.4] **Test Cases: Equivalence Partitioning** ⚡⚡
- [W10.5] **Test Cases: Boundary Value Analysis** ⚡⚡

Admin:

1. Submit weekly quiz (P)
2. ~~Take part in the CATcher load testing during the briefing on Mar 28th (P)~~

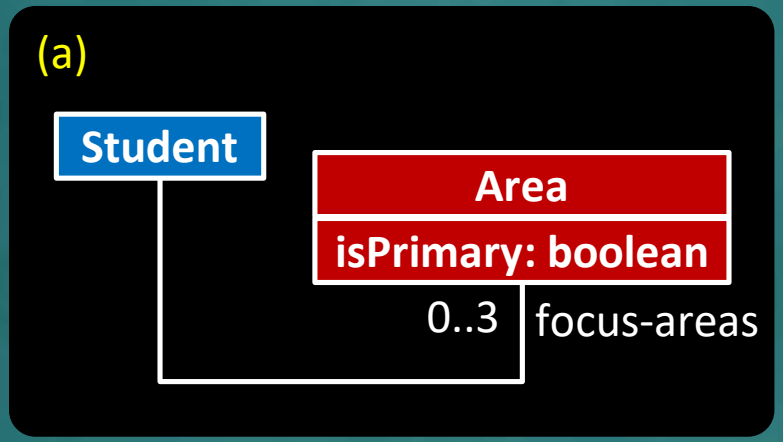
tP: v1.4

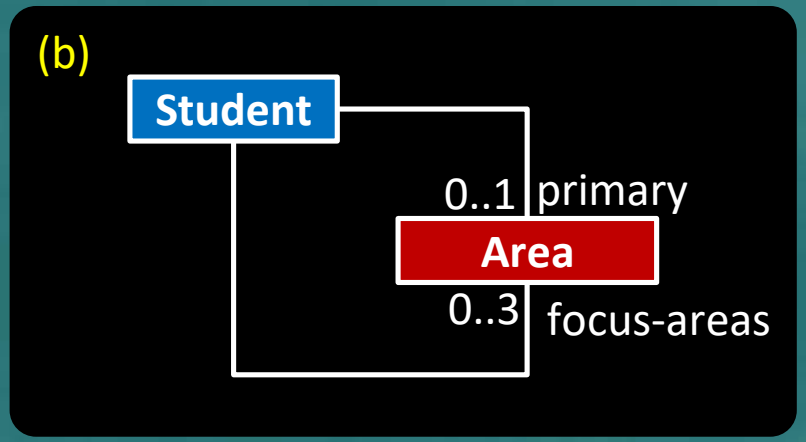
1. Do a postmortem of the previous iteration
2. Plan the alpha version (`v1.4`)
3. Deliver the alpha version (`v1.4`)
4. Smoke-test CATcher COMPULSORY (P)
5. Start updating UML diagrams in the DG

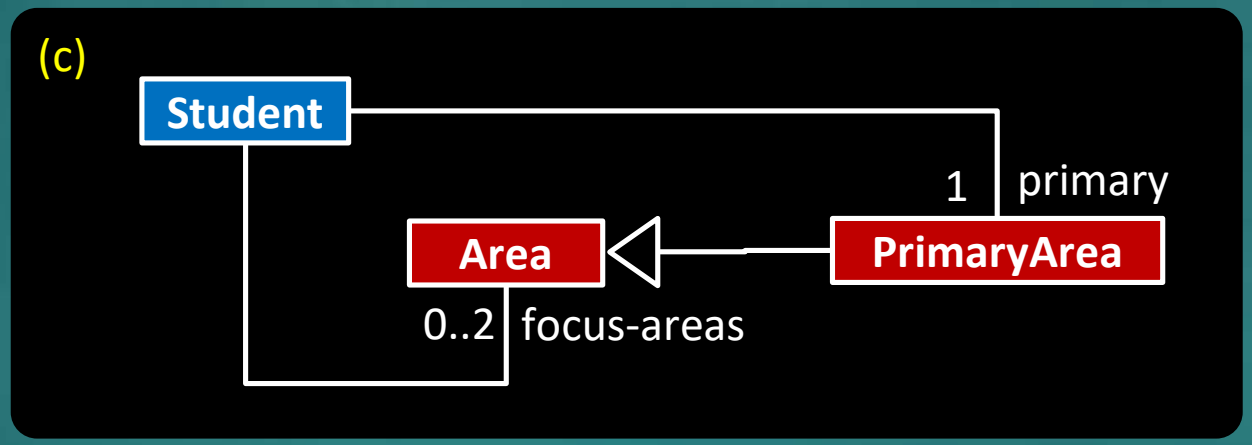

PR, review, merge
Team repo (remote)
GitHub
Forked repo on GitHub
Remember to 'protect' the master branch
push
pull
The feature branch workflow
branch
Local repo
You
Recommendation: switch only when you are already comfortable with the forking workflow

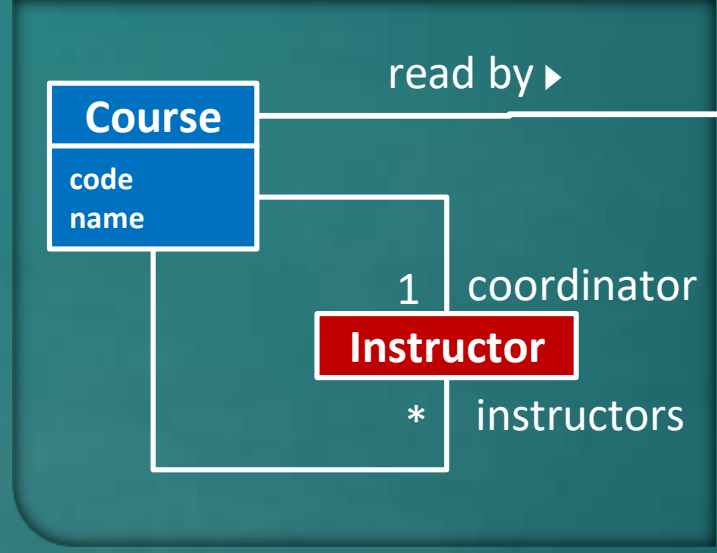
Choose the CCD that matches this description best.

Each student chooses up to 3 focus areas, one of which can be selected as the primary focus area.

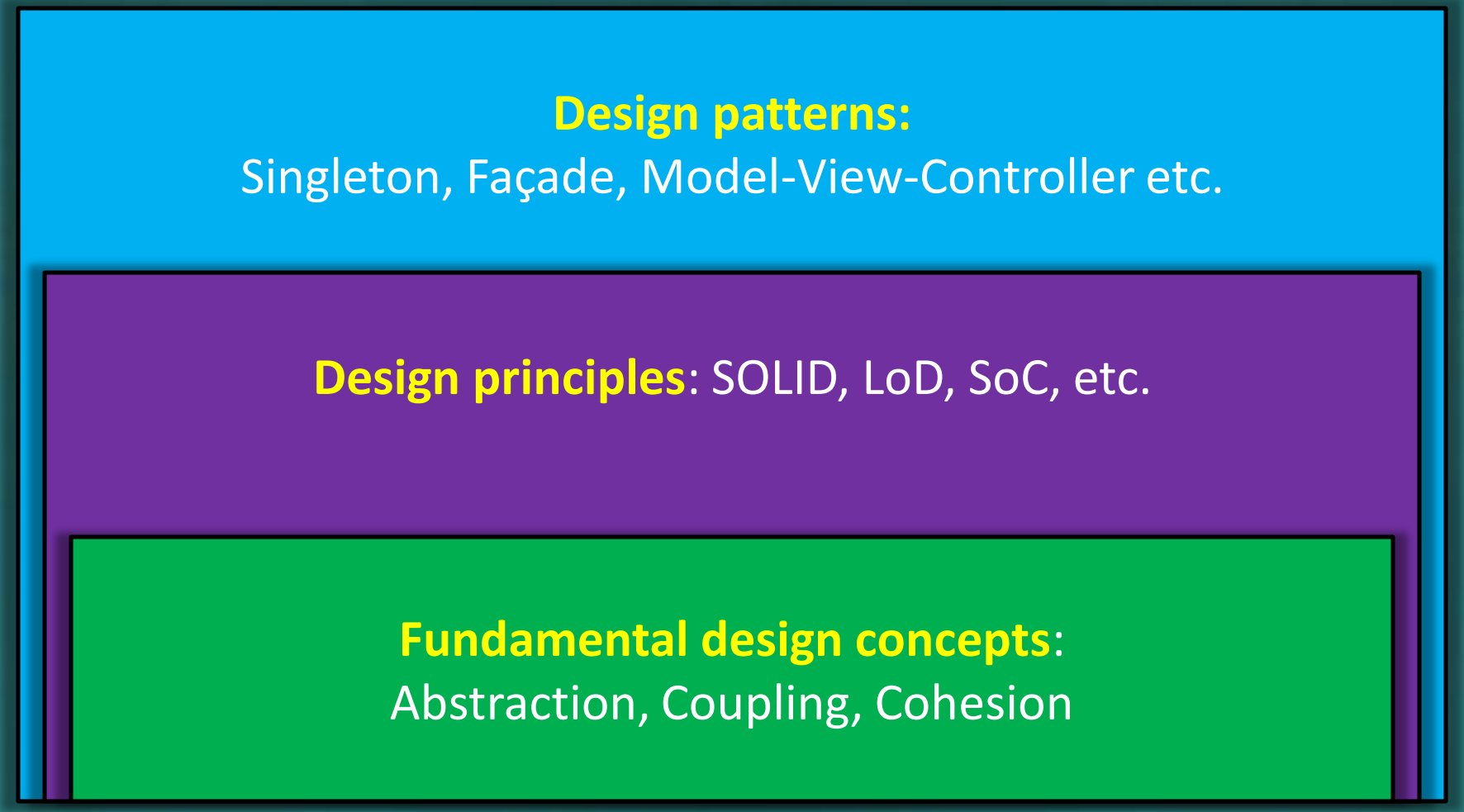








Note how the experience from the tutorial helped you 'pattern match' a solution here.


Design patterns:
Singleton, Façade, Model-View-Controller etc.
Design principles: SOLID, LoD, SoC, etc.
Fundamental design concepts:
Abstraction, Coupling, Cohesion

Week 10

Topics:

- [W10.1] **Design Patterns** ⚡⚡
- [W10.2] **Defensive Programming** ⚡⚡
- [W10.3] **Test Cases: Intro** ⚡⚡
- [W10.4] **Test Cases: Equivalence Partitioning** ⚡⚡
- [W10.5] **Test Cases: Boundary Value Analysis** ⚡⚡

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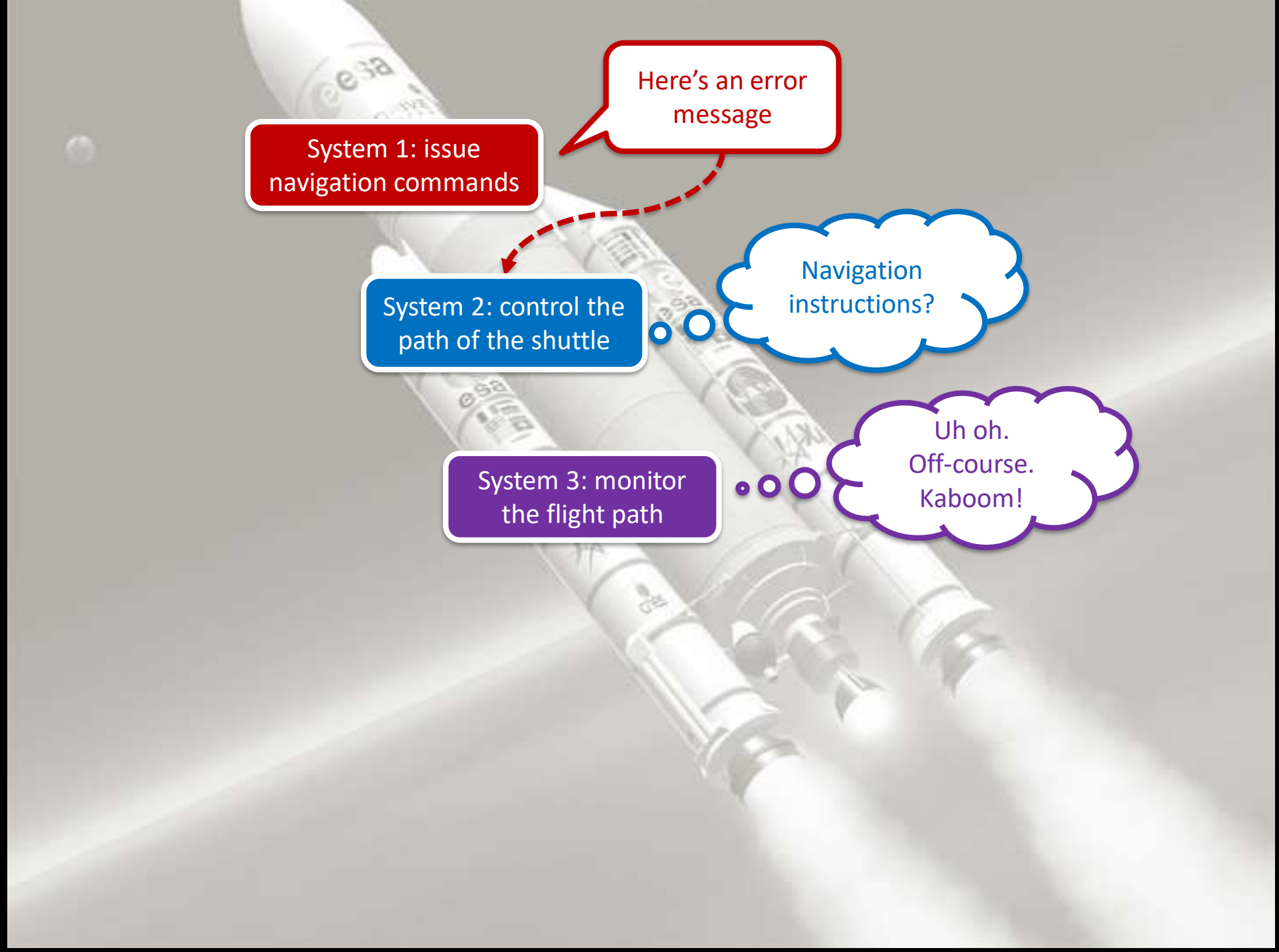
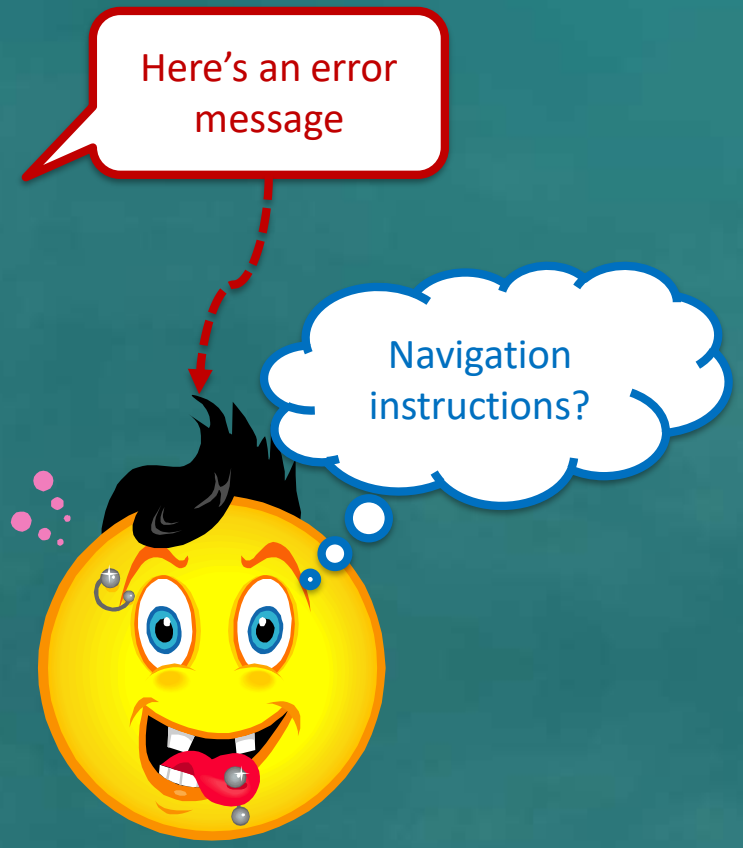
ARIANE 5 launch


Here's an error message
System 1: issue navigation commands
System 2: control the path of the shuttle
Navigation instructions?
System 3: monitor the flight path
Uh oh. Off-course. Kaboom!

Your code



Your teammate's code

Your code

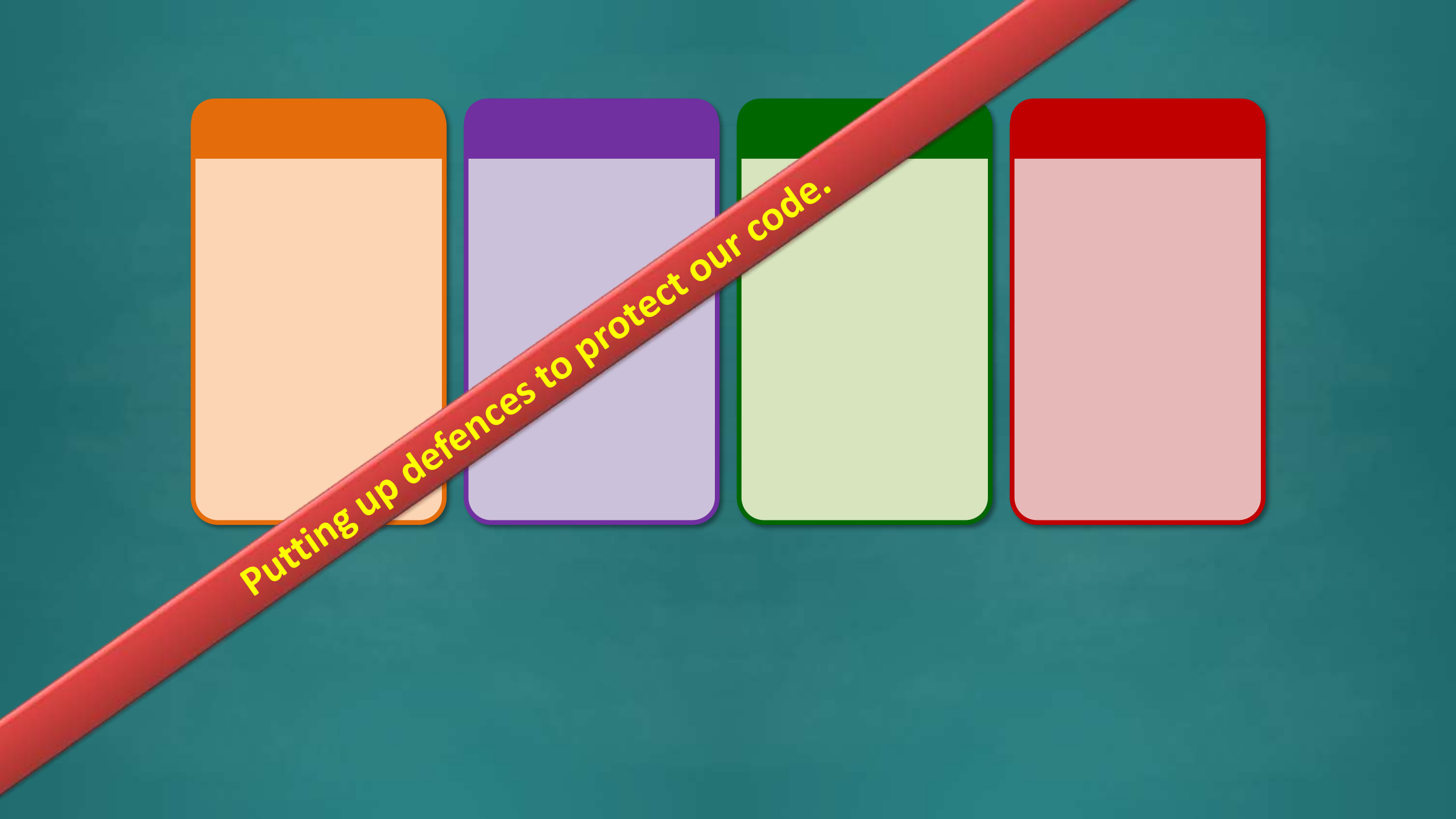
Putting up defences to protect our code.

Your teammate's code

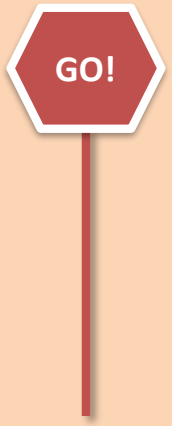
Your code

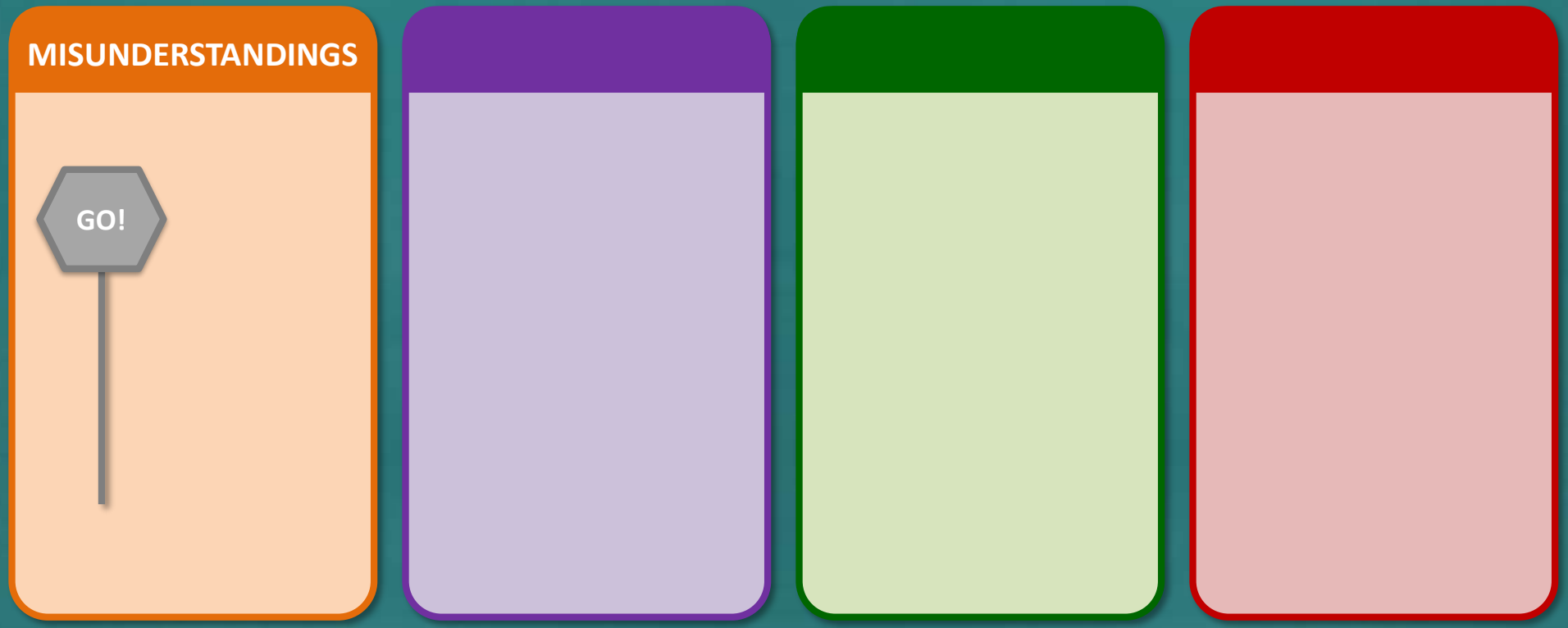
Putting up defences to protect our code.

Your teammate's code


Putting up defences to protect our code.

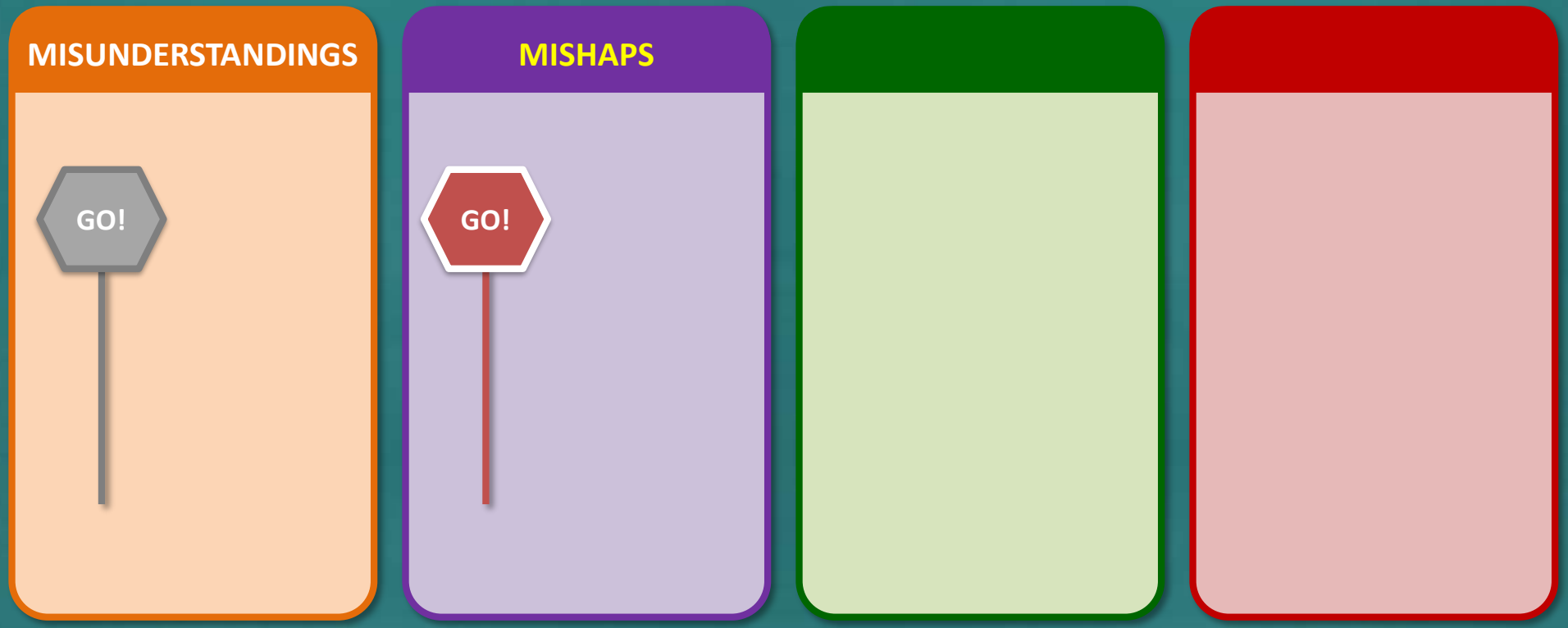
MISUNDERSTANDINGS



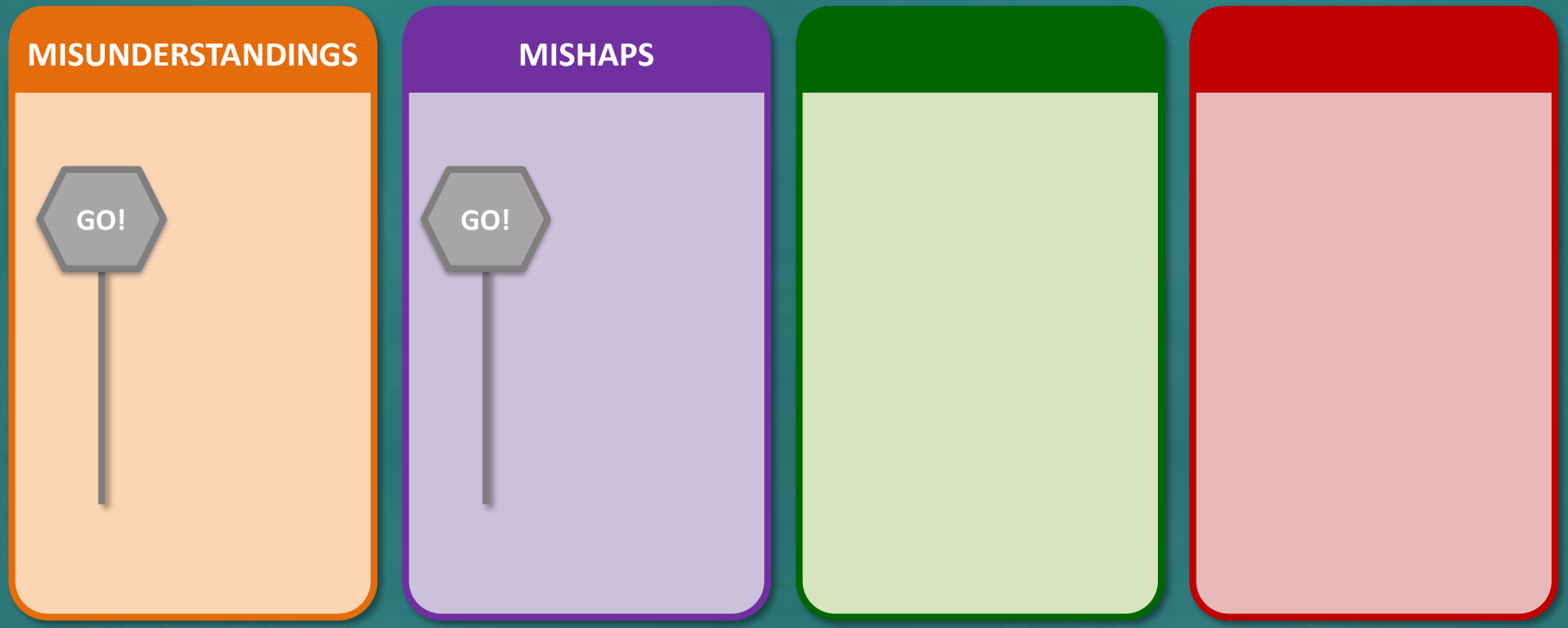


Assertions

```
x = getX();
assert x == 0 : "x should be 0";
...
```



Assertions



Assertions

Exceptions

Assertion or exception?

A developer screwed up

e.g., a method is given null as a parameter although null values are not supposed to reach that method

→ Use assertions to *abort*

A user screwed up (or a problem in the environment)

e.g., A file is not found in the expected location

→ Use exceptions to *handle*

Assertion or exception?

A developer screwed up
→ Use assertions to *abort*

A user screwed up
→ Use exceptions to *handle*

Consider this method:

```
execute(String userCommand)
```

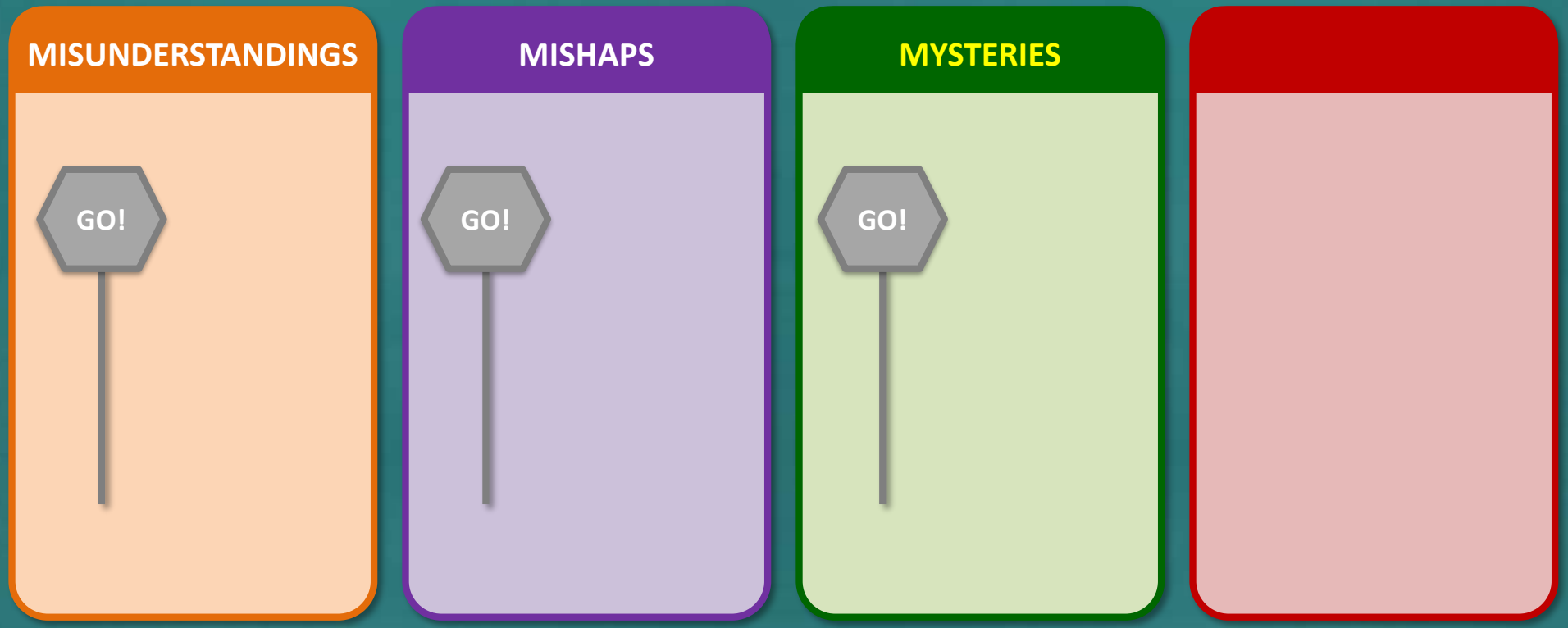
If the `userCommand` is in an incorrect format, it should be handled using,

- ○ Assertions
- ○ Exceptions
- ○ It depends

Q

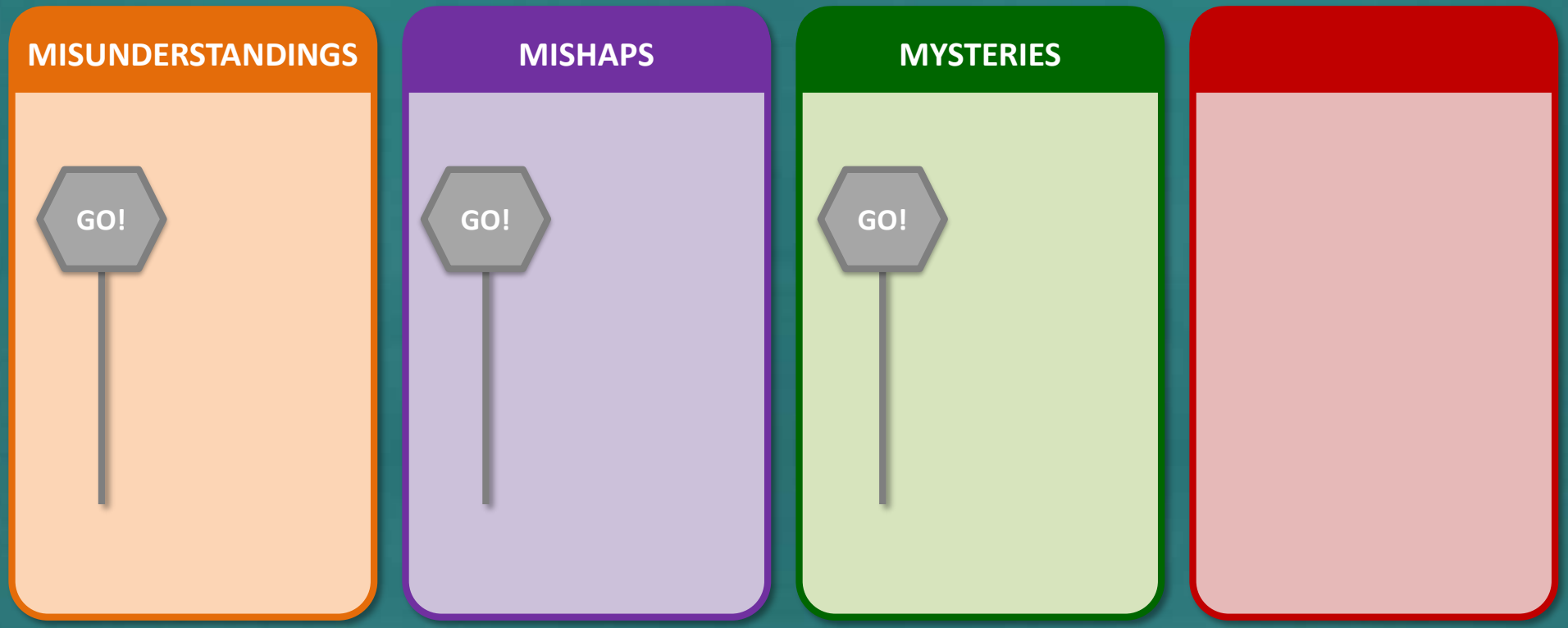
Assume your app tries to contact a Google API. If the **Google servers are down**, it should be handled using,

- ○ Assertions
- ○ Exceptions
- ○ It depends



Assertions

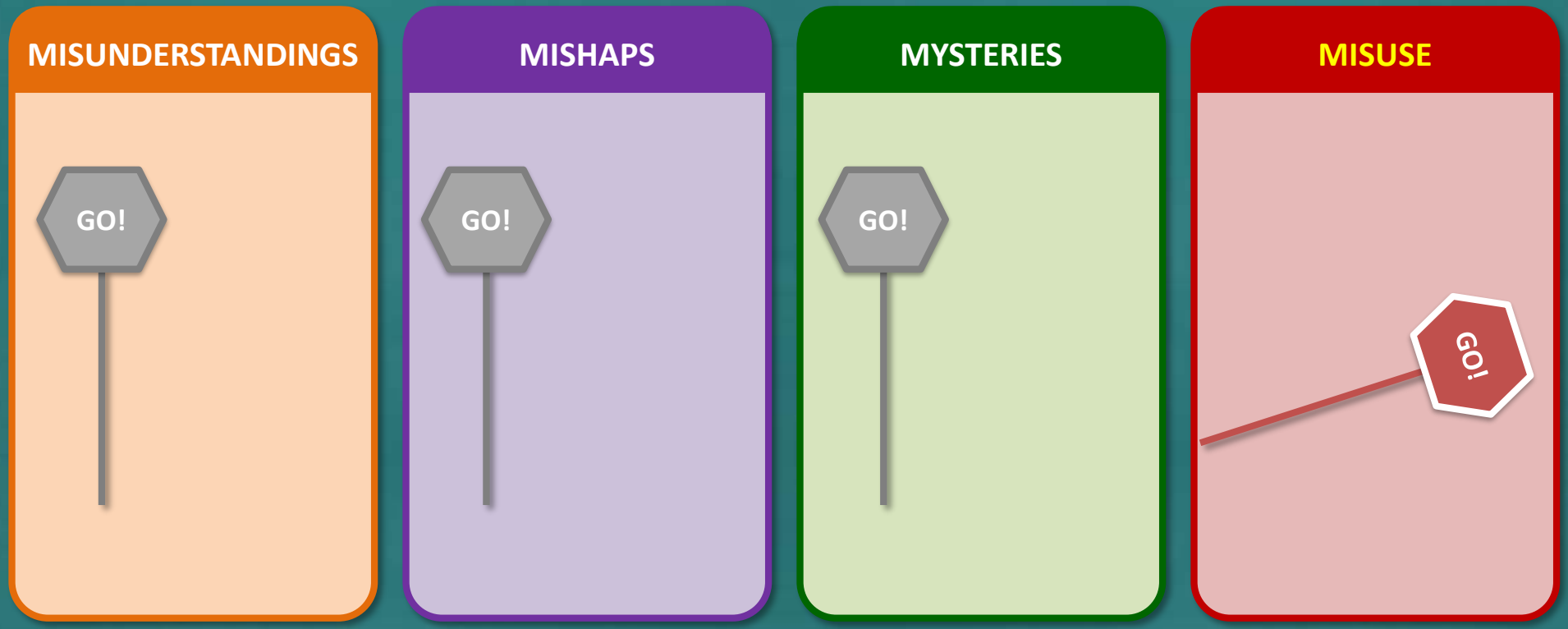
Exceptions



Assertions

Exceptions

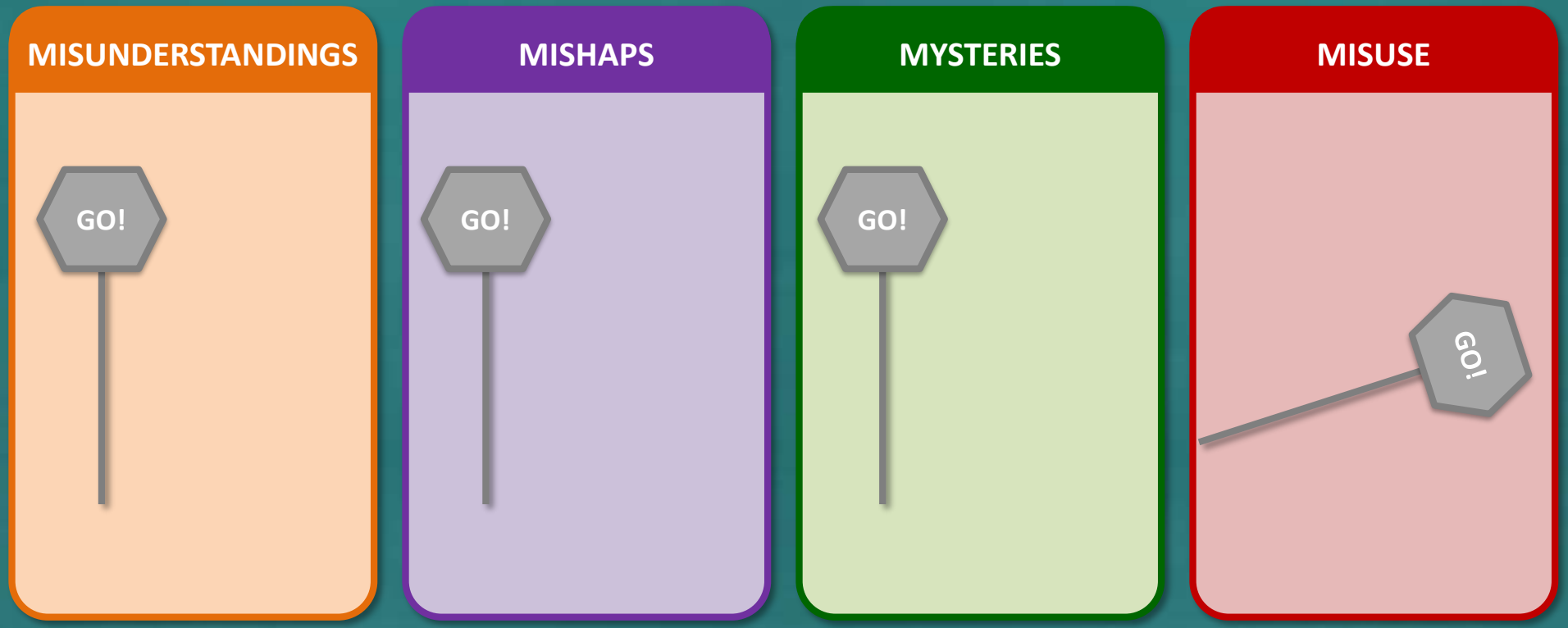
Logging



Assertions

Exceptions

Logging



Assertions

Exceptions

Logging

Defensive

Other three plus more!

Use all four in your own tP code!

Week 10

Topics:

- [W10.1] **Design Patterns** ⚡⚡
- [W10.2] **Defensive Programming** ⚡⚡
- [W10.3] **Test Cases: Intro** ⚡⚡
- [W10.4] **Test Cases: Equivalence Partitioning** ⚡⚡
- [W10.5] **Test Cases: Boundary Value Analysis** ⚡⚡

Admin:

1. Submit weekly quiz (P)
2. ~~Take part in the CATcher load testing during the briefing on Mar 28th (P)~~

tP: v1.4

1. Do a postmortem of the previous iteration
2. Plan the alpha version (`v1.4`)
3. Deliver the alpha version (`v1.4`)
4. Smoke-test CATcher COMPULSORY (P)
5. Start updating UML diagrams in the DG

Poll: of these three, which career path do your prefer?

a. System analyst (i.e., talk to users/customers)
b. Software engineer
c. Software tester

[don’t worry, there are other career paths too…]

Testing as a career…?

Easier to get in, get ahead

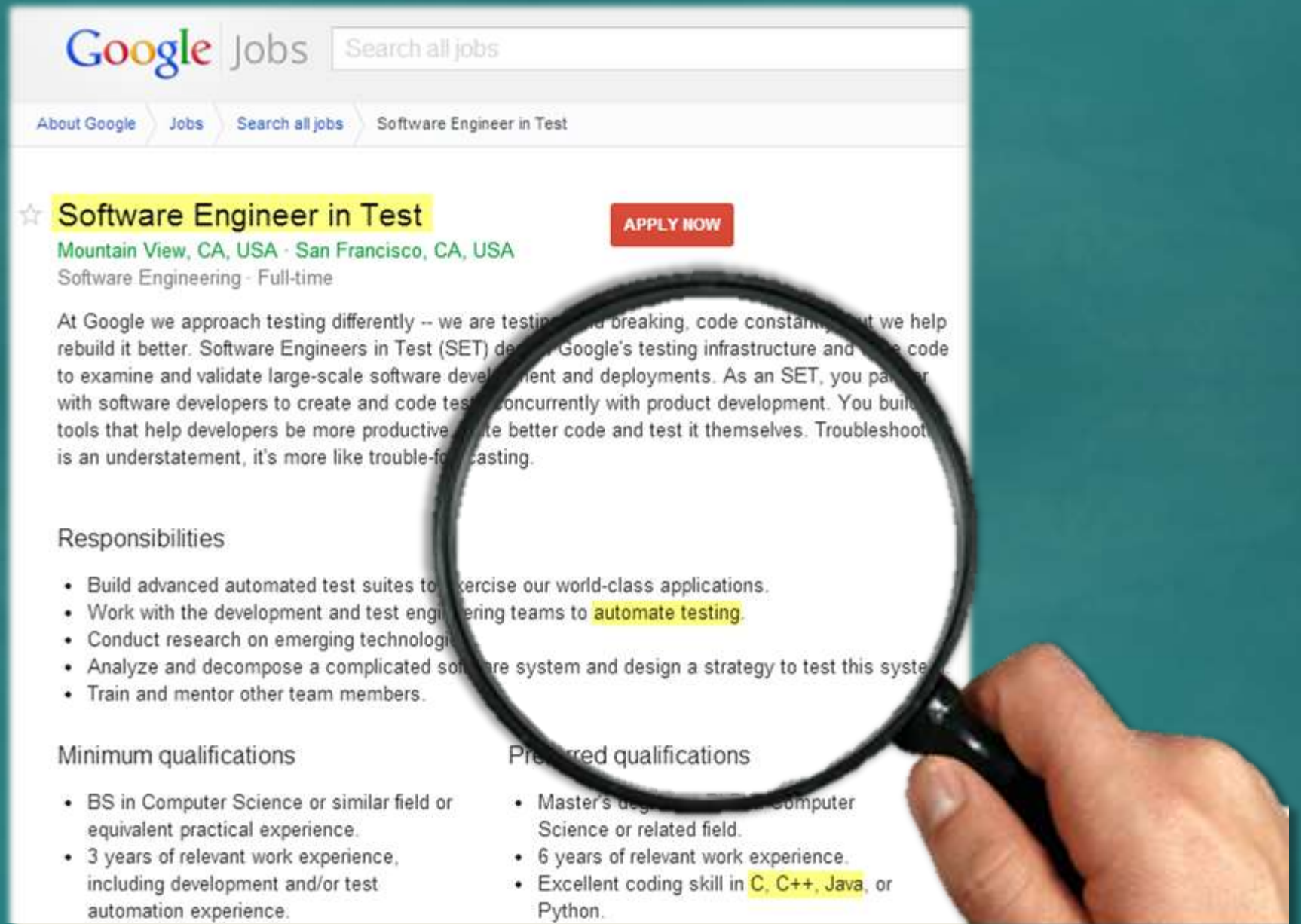
Blame others, not get blamed

Still can code

Testing as a career…?

So, don't rule out testing as a career path!

Testing as a career…?



:(

Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

41% complete

For more information about this issue and possible fixes, tps://www.windows.com/stopcode

If you call a support person, give them this info:

Stop code: CRITICAL_PROCESS_DIED

No need to celebrate!
This is a simulation ☺

Week 10

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Add all target features in v1.4 !